What Makes It Balance? Reflections

Name: _____ Period: _____

Decide together, as a group, where each equation belongs. Then each person writes it in their own box.

One solution	Infinite solutions	No solutions
Balanced for exactly one value	Always balanced	Never balanced

1 AFTER #1–7

Look at the equations in your "No solutions" column. What is the same about the two sides? What is different?

2 AFTER #7

If the equation is balanced no matter what value of x we choose on the slider, then ANY value seems to balance the equation. How many possible solutions are there for these types of equations?

3 AFTER #9

Equation 9 has a solution of 0. Explain why "the solution is 0" is not the same as "there is no solution."

4 AFTER #8–10

Look at the blue terms in each of your three columns. What do you notice?

5a AFTER #11–12

The slider broke. How can you tell just by looking that an equation has only one solution?

5b **How can you tell just by looking that an equation has infinite solutions?**

5c **How can you tell just by looking that an equation has no solutions?**